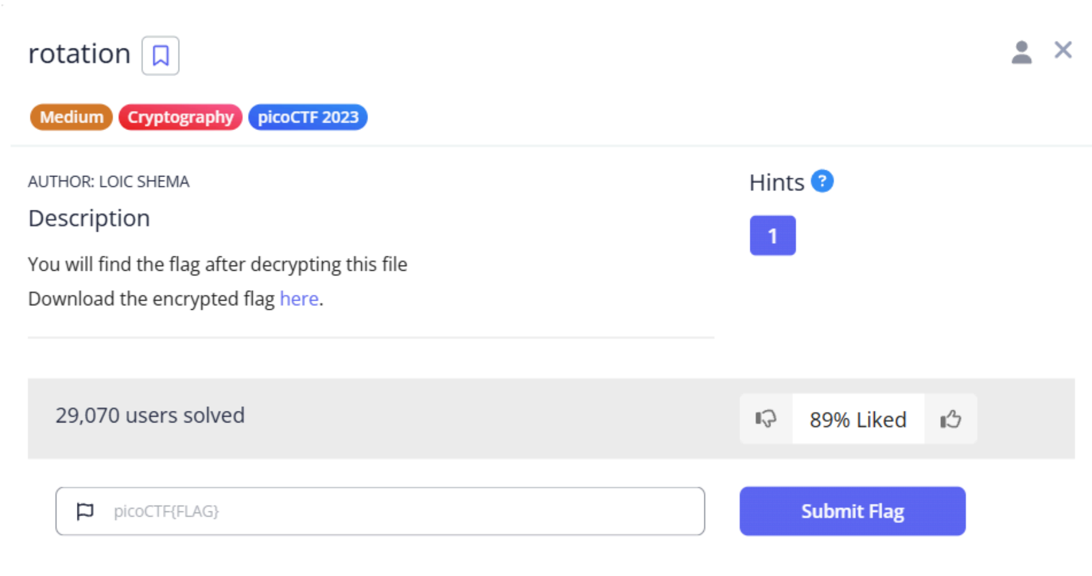
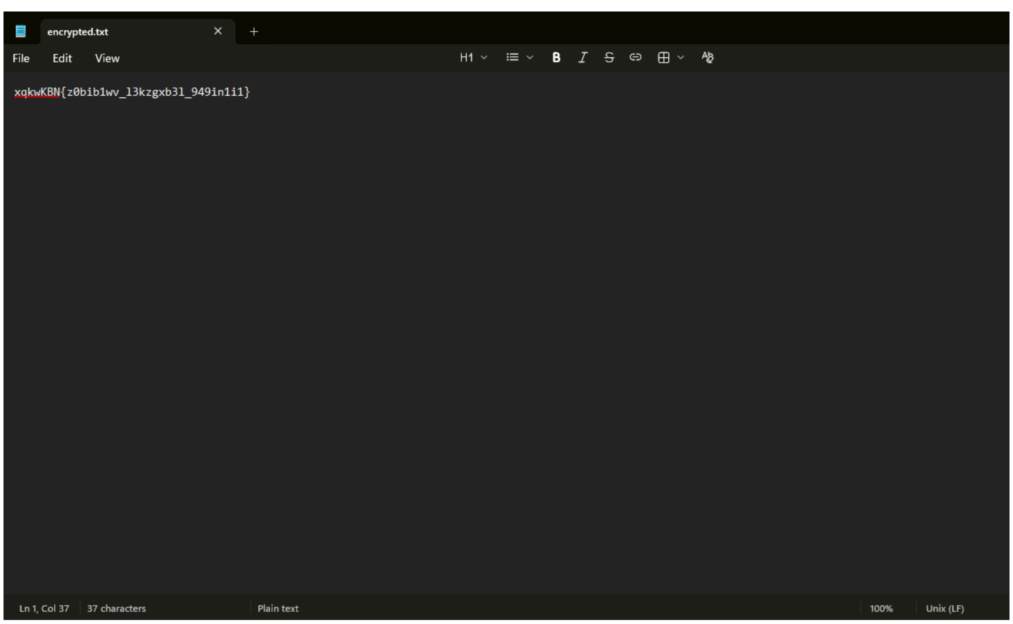
HERNANDO, PANTINE B. **BSIT-3**

CHALLENGE - 10 - CRYPTOGRAPHY



The **Rotation** challenge in picoCTF 2023 is a beginner-friendly cryptography task that involves decoding a message encrypted using a rotation cipher (also known as a Caesar cipher).



Objective

The goal is to decrypt the provided encoded file and retrieve the flag in the format:
picoCTF{r0tat1on_d3crypt3d_949af1a1}

Approach

Upon downloading the encrypted file, the contents appear as unreadable text. This suggests the use of a simple substitution cipher, specifically a rotation cipher where each letter is shifted by a fixed number in the alphabet.

Since the rotation value is unknown, the most effective method is to try all possible shifts (ROT1 to ROT25). This is considered a brute-force approach in basic cryptography.

Solution Process

1. Open the encrypted file using a text editor.
2. Copy the ciphertext.
3. Use a rotation/caesar cipher tool (or script) to test all 25 possible shifts.
4. Scan the outputs for readable English text.
5. Identify the correct shift when the message becomes meaningful.
6. Locate the flag within the decoded text.

Key Insight

One of the rotations will clearly reveal a readable sentence containing the flag. The correct rotation stands out because it produces coherent words and follows the expected flag format.

Conclusion

This challenge demonstrates how weak simple encryption methods like rotation ciphers are, especially when brute-force techniques are applied. It also reinforces foundational skills in cryptography, such as pattern recognition and systematic testing.